

AIHUI, China

WMO: 504680

Lat: 50.2481N

Lon: 127.4619E

Elev: 166

StdP: 99.35

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.1	-30.9	-36.3	0.1	-32.6	-34.4	0.1	-30.4	8.0	-15.3	7.2	-16.8	1.4	320	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	30.9	20.5	29.1	20.1	27.5	19.6	23.1	27.7	22.1	26.5	21.2	25.4	3.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.7	16.7	25.2	20.7	15.7	24.4	19.8	14.8	23.5	69.6	27.9	66.0	26.5	62.6	25.3	26.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.2	7.0	6.1	-34.8	34.1	2.7	1.8	-36.8	35.4	-38.4	36.5	-39.9	37.5	-41.9	38.8		
			-34.9	24.7	2.7	1.3	-36.9	25.6	-38.5	26.4	-40.0	27.1	-41.9	28.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	1.5	-21.1	-16.6	-7.3	4.2	12.5	18.9	21.7	19.4	12.5	2.8	-10.5	-20.2	(6)
(7)		DBStd	16.00	4.78	4.98	6.10	4.54	4.20	3.52	2.58	3.02	4.10	4.85	6.72	4.60	(7)
(8)		HDD10.0	4248	965	744	536	182	21	0	0	1	21	228	614	936	(8)
(9)		HDD18.3	6387	1223	978	794	424	187	35	4	23	178	482	864	1194	(9)
(10)		CDD10.0	1128	0	0	0	9	98	267	362	291	97	4	0	0	(10)
(11)		CDD18.3	226	0	0	0	1	7	51	108	56	4	0	0	0	(11)
(12)		CDH23.3	1943	0	0	0	9	128	538	807	408	51	1	0	0	(12)
(13)		CDH26.7	538	0	0	0	2	34	177	230	91	5	0	0	0	(13)
(14)	Wind	WSAvg	2.5	2.3	2.4	3.0	3.3	3.1	2.4	2.2	2.1	2.4	2.7	2.4	2.2	(14)
(15)	Precipitation	PrecAvg	511	2	3	6	24	44	84	130	124	61	20	8	5	(15)
(16)		PrecMax	786	6	11	19	126	152	144	308	244	118	68	28	13	(16)
(17)		PrecMin	344	0	0	1	3	6	19	15	35	11	0	0	0	(17)
(18)		PrecStd	101	2	3	5	23	31	36	74	50	32	15	7	3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-7.1	0.7	11.3	24.4	30.2	33.8	33.6	31.0	27.2	20.1	8.6	-6.1	(19)
(20)			MCWB	-9.0	-2.9	3.9	10.9	15.1	20.0	22.2	21.7	18.2	10.6	3.7	-7.3	(20)
(21)		2%	DB	-10.1	-2.9	7.7	18.9	26.9	30.8	31.0	29.0	24.8	15.8	4.4	-8.8	(21)
(22)			MCWB	-11.6	-5.5	1.9	8.5	14.2	19.4	21.9	21.4	16.7	8.9	0.8	-9.9	(22)
(23)		5%	DB	-12.1	-6.1	4.6	15.6	23.9	28.5	29.2	27.4	22.5	13.0	1.4	-11.4	(23)
(24)			MCWB	-13.4	-8.2	0.0	7.0	12.8	18.4	21.4	20.5	15.5	7.1	-1.2	-12.5	(24)
(25)		10%	DB	-14.1	-8.7	2.1	12.9	21.3	26.4	27.6	25.7	20.2	10.5	-1.1	-13.4	(25)
(26)			MCWB	-15.1	-10.4	-1.5	5.6	11.8	17.9	20.8	19.7	14.2	5.6	-3.0	-14.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-9.0	-2.7	4.5	12.0	16.7	22.9	24.7	24.0	20.0	12.6	3.7	-7.2	(27)
(28)			MCDWB	-7.5	0.2	10.3	22.1	25.1	30.1	29.5	28.4	24.5	17.9	7.9	-6.0	(28)
(29)		2%	WB	-11.6	-5.3	2.2	9.3	15.2	21.1	23.4	22.6	18.1	9.6	0.9	-10.0	(29)
(30)			MCDWB	-10.2	-3.1	7.2	17.6	24.0	27.5	28.1	26.8	22.4	14.2	4.1	-8.8	(30)
(31)		5%	WB	-13.3	-8.1	0.2	7.5	14.1	19.9	22.6	21.6	16.7	7.7	-0.8	-12.4	(31)
(32)			MCDWB	-12.2	-6.2	4.1	14.6	21.7	25.6	26.9	25.6	21.0	12.2	1.2	-11.4	(32)
(33)		10%	WB	-15.1	-10.3	-1.3	5.9	12.8	18.8	21.8	20.5	15.2	6.0	-3.0	-14.3	(33)
(34)			MCDWB	-14.1	-8.7	2.0	11.9	19.6	24.6	25.9	24.3	18.9	10.1	-1.1	-13.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.9	11.7	11.5	11.5	12.1	11.1	9.4	9.8	11.7	10.4	9.4	8.8	(35)
(36)			MCDBR	10.9	13.1	13.0	16.6	16.9	15.2	12.2	12.6	15.0	14.6	11.3	9.7	(36)
(37)			MCWBR	9.8	10.9	8.5	9.0	7.3	6.0	5.0	5.7	7.5	8.5	8.3	8.6	(37)
(38)		5% WB	MCDBR	10.8	13.1	12.1	15.2	14.3	12.1	9.7	10.2	12.4	13.0	11.0	9.6	(38)
(39)			MCWBR	9.7	11.0	8.2	8.5	6.7	5.6	4.8	5.4	6.9	8.2	8.2	8.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.272	0.302	0.374	0.507	0.566	0.501	0.484	0.431	0.393	0.387	0.319	0.279	(40)	
(41)		taud	2.161	2.090	1.955	1.791	1.733	2.003	2.107	2.240	2.263	2.117	2.151	2.118	(41)	
(42)		Ebn at Noon	750	815	806	732	708	767	775	801	791	700	670	667	(42)	
(43)		Edh at Noon	76	107	148	197	219	168	149	125	109	105	78	69	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.68	2.88	4.42	5.00	5.43	5.84	5.41	4.81	3.87	2.57	1.66	1.30	(44)	
(45)		RadStd	0.06	0.10	0.28	0.53	0.45	0.74	0.46	0.34	0.32	0.15	0.10	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page